

## Skills Assessed

The main goal of this lab is to assess skills that students have learned throughout the quarter and apply them to a real-world dataset. This lab asks students to use R and tidyverse functions such as `mutate()`, `filter()`, `count()`, `map()`, and `arrange()`. Students are also required to write their own functions and include conditional statements within those functions.

Another important part of data analysis is being able to create visual representations of data. Because of this, the lab asks students to create several types of charts, including scatterplots and bar charts. The lab also assesses students' ability to work with statistical models by asking them to create and interpret linear regression models.

## Easy Problems

Question 1 should not be too challenging for students. The solution requires students to count observations by discovery method and create a bar chart to visualize the results. These are skills students learned earlier in the quarter and have continued practicing throughout the course.

Questions 2 and 5 can also be considered easier questions. They do require students to create functions, which is a topic covered later in the class, but the functions themselves are not meant to be very difficult. In both questions, students create a simple function that takes one input and uses conditional statements inside the function body.

## Hard Problems

Question 9 will likely be challenging for many students. It asks students to create a function with multiple parameters, including a data frame. The function also needs to be reusable later in the question, which makes it more complex. In addition, students have to create a data frame inside the function, making this question more difficult than the earlier function questions.

Question 11 will likely be the most challenging problem in the lab. This question uses simulation, which was one of the topics covered toward the end of the course. It assesses several skills at once, including random sampling, repeated simulations, data visualization, and comparing simulated results.

## Outside Resources

<https://www.codecademy.com/learn/learn-linear-regression-in-r/modules/linear-regression-in-r/cheatsheet>

<https://numiqo.com/tutorial/linear-regression>

## Additional Source Explanation

The main dataset source for this lab was the NASA Exoplanet Archive Planetary Systems Composite Parameters table. This source was used because it provides a real dataset about confirmed exoplanets, rather than a built-in R dataset or a dataset already used in class. The NASA Exoplanet Archive column definitions page was especially important because it explained variables such as `discoverymethod`, `disc_year`, `pl_rade`, `pl_bmasse`, `pl_orbper`, `pl_eqt`, `sy_dist`, `sy_pnum`, `sy_snum`, `st_teff`, and `st_mass`.

I used the data dictionary to decide which variables could support clear lab questions. For example, `disc_year` led to a question about discoveries over time, `pl_rade` and `pl_bmasse` led to questions about planet size and mass, and `sy_dist` led to questions about distance from Earth.

I also used NASA's explanation of exoplanet detection methods when thinking about questions involving `discoverymethod` and `pl_orbper`. Since different methods are better at detecting different kinds of planets, it made sense to ask students to compare discovery methods by counts, distance, and orbital period. NASA's habitable zone resources helped support the rocky-zone candidate question. That question is intentionally written as a simplified classroom classification, not as a scientific claim that a planet is truly habitable.

Class materials were also used when deciding what skills the questions should assess. The early questions connect to Week 2 and Week 4 skills such as `ggplot` visualizations, `count()`, `filter()`, `group_by()`, `summarize()`, `median()`, and linear regression interpretation. The later questions connect more directly to Week 7 and Week 8 material. Week 7 focused on writing functions and using `map()` family functions to avoid repeated code. Week 8 focused on data-frame functions, tidy evaluation, checking data quality, and simulations. Questions 8, 9, 10, 11, and 12 were designed to make sure the lab included enough Week 7 and later content.

## References

NASA Exoplanet Archive. Planetary Systems Composite Parameters.

[https://exoplanetarchive.ipac.caltech.edu/cgi-bin/TblView/nph-tblView?app=ExoTbls&config=P\\_SCompPars](https://exoplanetarchive.ipac.caltech.edu/cgi-bin/TblView/nph-tblView?app=ExoTbls&config=P_SCompPars)

NASA Exoplanet Archive. Planetary Systems and Planetary Systems Composite Parameters Column Definitions.

[https://exoplanetarchive.ipac.caltech.edu/docs/API\\_PS\\_columns.html](https://exoplanetarchive.ipac.caltech.edu/docs/API_PS_columns.html)

NASA Science. How We Find and Characterize Exoplanets.  
<https://science.nasa.gov/exoplanets/how-we-find-and-characterize/>

NASA Science. What Is the Habitable Zone?  
<https://science.nasa.gov/exoplanets/what-is-the-habitable-zone-or-goldilocks-zone/>

Tidyverse Documentation.  
<https://www.tidyverse.org/packages/>

dplyr Documentation.  
<https://dplyr.tidyverse.org/>

ggplot2 Documentation.  
<https://ggplot2.tidyverse.org/>

purrr Documentation.  
<https://purrr.tidyverse.org/>

R Project. R Documentation and Manuals.  
<https://www.r-project.org/other-docs.html>

Codecademy. Linear Regression in R Cheatsheet.  
<https://www.codecademy.com/learn/learn-linear-regression-in-r/modules/linear-regression-in-r/cheatsheet>

Numiqo. Linear Regression Tutorial.  
<https://numiqo.com/tutorial/linear-regression>

Generative AI Use

I used AI on this project. I used it to check mainly whether the lab addressed the project rubric, review wording on some of the questions and check whether the code was all set. I also used to help me identify where sources should be included.

<https://chatgpt.com/share/6a23c339-4d34-83e8-b8c7-628e4a7bb54e>